Carisurg

Assignment 15

Interim Documentation

Due Date: July 4th, 2026.

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Feasibility Memo

Verdict

Proceed to a baseline, with caveats. The extract carries a genuine, complete triage label (ESI) and clinically sensible vitals for 55,121 encounters — enough to build and evaluate a Week 6 baseline — but it is a single-site, non-Caribbean, US academic-hospital dataset, so any result is a proof-of-concept for the method, not evidence that a model would work at Mercer.

Dataset Summary

The extract holds 55,121 ED encounters across 225 columns: 25 structured fields (triage vitals, demographics, arrival details) plus 200 chief-complaint flags. All records come from three departments (A, B, C) inside one hospital system.

1. **Age**: mean 55.3 years, range 18–107 (adult-only ED).
2. **Gender**: 57.6% female, 42.4% male.
3. **Race**: 53.4% White, 29.0% Black, 16.4% Other/Refused/Unknown/Asian/AI-AN/NHPI combined.
4. **Ethnicity**: 81.9% Non-Hispanic, 17.9% Hispanic or Latino.
5. **Insurance**: Medicaid (38.9%) and Medicare (31.9%) together cover most encounters — a publicly-insured, likely lower-income population, which is a useful point of comparison when judging fit to Mercer's own patient mix.
6. **ESI** (Triage_Level) balance: 1 = 0.1%, 2 = 32.5%, 3 = 49.0%, 4 = 16.1%, 5 = 2.2%. As expected for an ED, most patients are mid-acuity and very few are ESI 1.

7. **Temperature** is recorded in °F (median 98.0°F), not °C.

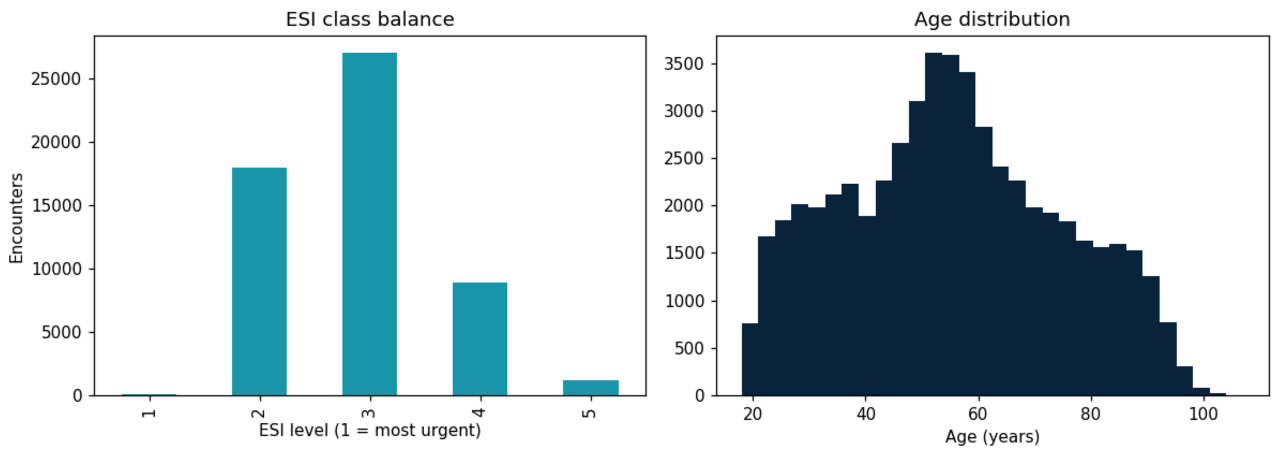

Figure 2 — ESI class balance and age distribution

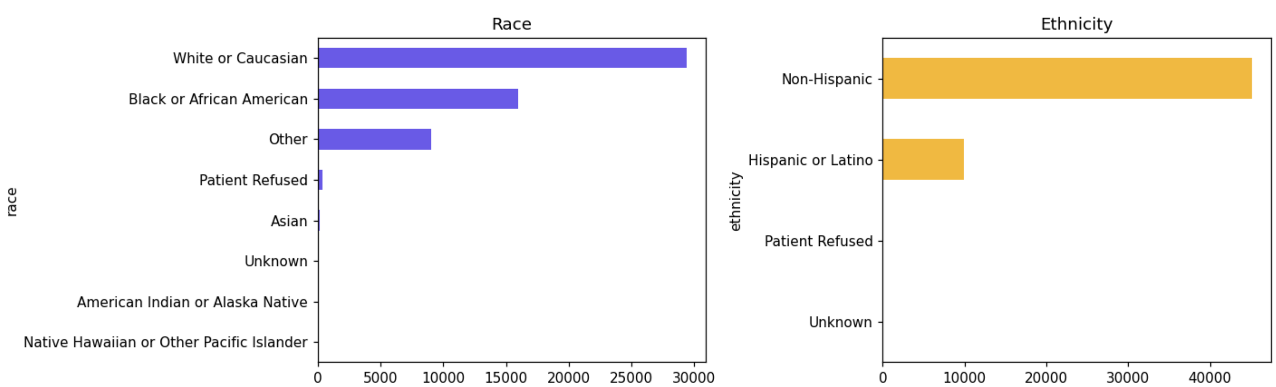

Figure 3 — Race and ethnicity

Top Three Concerns

1. **Outcome leakage:** Disposition and previous disposition (admit/discharge/transfer) are only known after the visit concludes. Used as model inputs, they would leak the answer, so both are excluded from any predictive feature set — a hard rule, not a judgment call.
2. **Single-site, non-Caribbean representativeness**: Every encounter comes from one US academic-hospital system with a publicly-insured, majority-White/Black patient mix (Figure 3). Mercer General's Caribbean catchment — case-mix, demographics, insurance structure, and resourcing — is materially different, and nothing in this extract tests whether a model trained on it would transfer. This is the single biggest reason the verdict carries caveats rather than being an unqualified "proceed."
3. **Sparse, heavy-tailed, and suspiciously complete data:** 149 of the 200 chief-complaint flags (74.5%) fire in fewer than 0.5% of encounters and carry little standalone signal (Figure 4, above). Glucose and temperature both have real long tails — 10.3% of glucose readings and 6.4% of temperature readings sit outside the interquartile fence (Figure 6, below) — most likely genuine ED pathology (hyperglycaemia, fever) rather than data errors, but worth a clinical sanity-check before Week 6. Separately, every structured column in this extract is 100% complete — no missing ESI labels, no missing vitals anywhere (Figure 1, committed in /figs). That is unusual for a raw EHR pull and is flagged as a caveat, not a strength: it likely reflects cleaning done by the dataset's original publisher, so we cannot confirm from this extract alone how a live Mercer capture tool would actually behave when a nurse skips a field under time pressure.

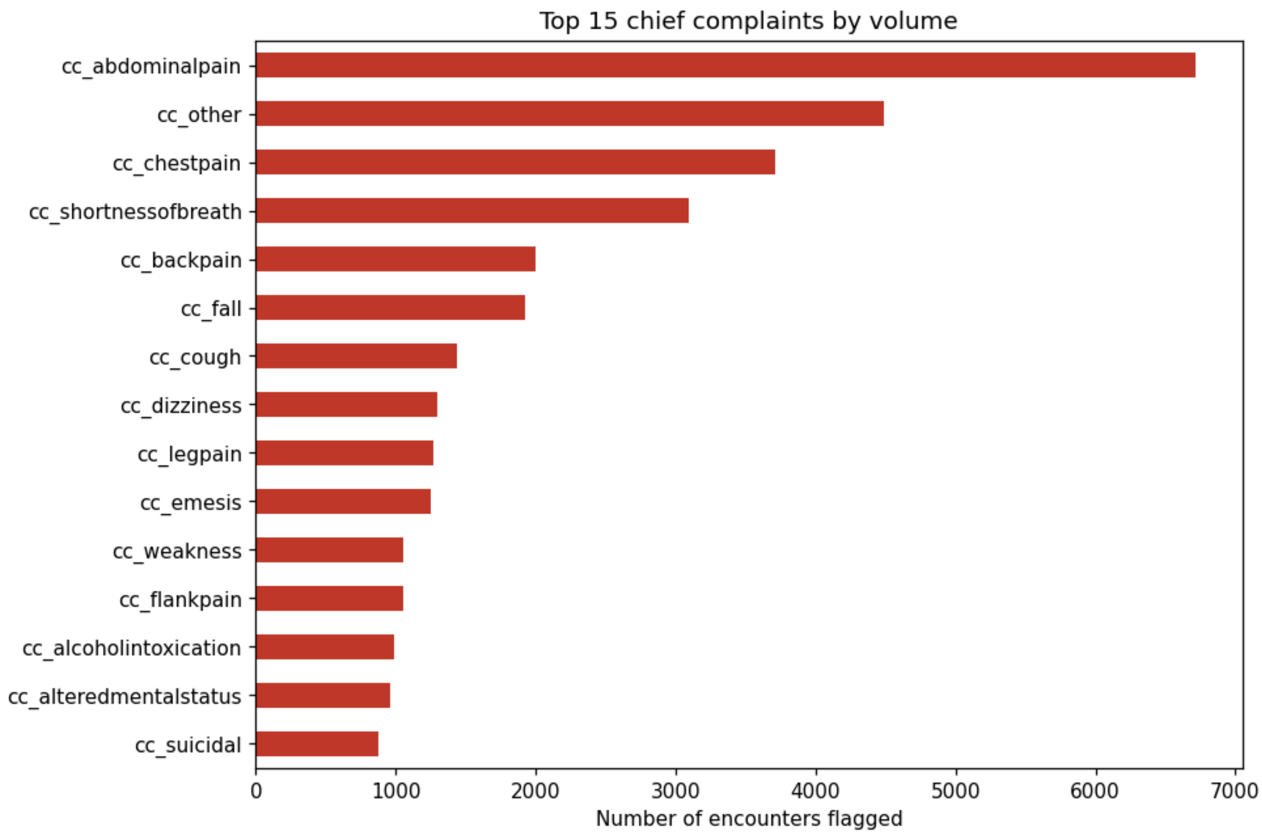

Figure 4 — Top 15 chief complaints by volume

Top Three Reasons to Proceed

1. **A real, complete triage label.** Every one of the 55,121 encounters carries a genuine ESI level 1–5 (Figure 2) — we can learn the actual target, not a stand-in, and there are zero rows to discard for a missing label.
2. **Vitals separate acuity in the clinically expected direction.** Oxygen saturation is the single strongest vital–ESI relationship ($r = 0.18$): lower SpO_2 tracks with more urgent (lower) ESI levels, exactly as a triage nurse would expect (Figure 6, below; Figure 5 committed in /figs). Respiratory rate and heart rate follow the same expected direction ($r = -0.10$ each). Age is the strongest individual signal found in this profiling pass ($r = -0.24$): older patients skew toward more urgent triage, consistent with reduced physiological reserve.
3. **A documented, reproducible pipeline.** Every cleaning decision — flagging physiologically impossible values (4 respiratory-rate readings, 25 glucose readings), median imputation of vitals, category fill for text fields — is logged in clean_triage() and can be re-run end-to-end from the raw CSV by any auditor.

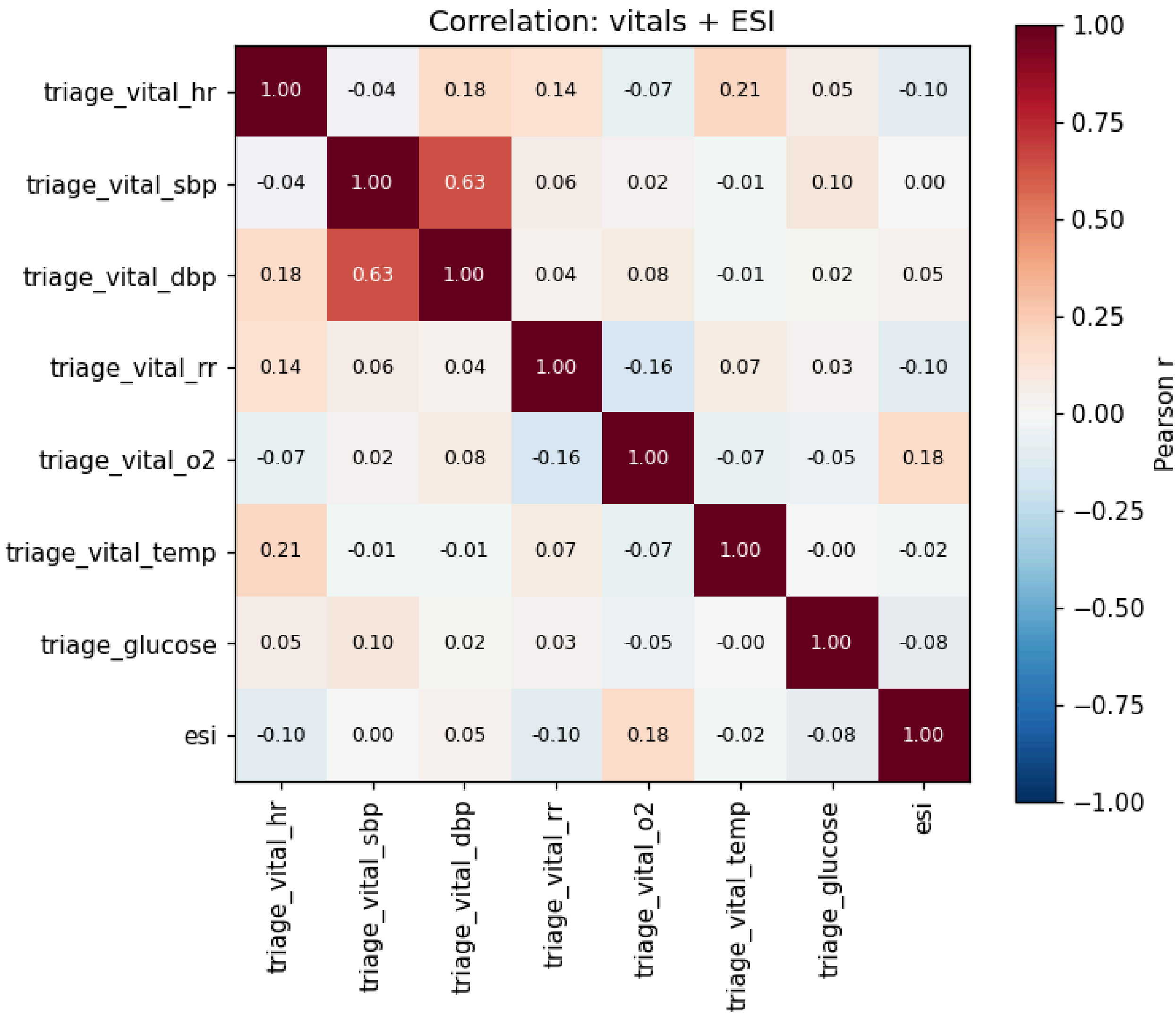

Figure 6 — Correlation matrix, vitals and ESI

Caveats

1. **Temperature is in °F, not °C** — any comparison to Celsius-based clinical thresholds must convert first.
2. **Chief-complaint flags are sparse**; 74.5% are near-constant and carry little signal alone.
3. **Correlations reported here are association only — not proof, and not a model**. The strongest single relationship found (age, $r = -0.24$) is still a weak-to-moderate linear association, and ESI is a multifactorial clinical judgement no single feature reproduces.
4. **External validity for a Caribbean ED (Mercer General) is completely untested**. This is a US, single-system, non-Caribbean dataset; a Week 6 baseline built on it demonstrates a method, not a Mercer-ready model.
5. **The extract's apparent 100% completeness should not be assumed for live Mercer data capture** — a real front-desk tool under a 3–5 minute triage window will very likely produce genuine missingness that this extract does not exhibit.

Top-10 Feature Shortlist

Ranked by combining clinical intuition (what a triage nurse would trust) with simple correlation against ESI in this extract. Leakage columns (disposition, previousdispo) are excluded on purpose.

#	Feature	r vs ESI	Why a triage nurse would trust it
1	age	−0.24	Strongest single signal found; reduced physiological reserve in older patients raises acuity for the same complaint.
2	triage_vital_o2 (SpO_2)	+0.18	Hypoxia is a first-look ABCs (Airway-Breathing-Circulation) marker; the strongest vital-ESI relationship.
3	cc_chestpain	−0.16	Classic ACS/PE rule-out complaint; strongest single chief-complaint correlation.
4	cc_shortnessofbreath	−0.15	Direct respiratory distress is a red flag, the second-strongest complaint correlation.
5	cc_suicidal	−0.14	Safety-critical; mandates rapid response regardless of vitals.
6	cc_alcoholintoxication	−0.14	High-frequency ED presentation with airway/safety risk.
7	cc_alteredmentalstatus	−0.13	Neuro red flag: altered mentation is high-acuity, almost by definition.
8	arrivalmode (ambulance flag)	descriptive	50.7% of ambulance arrivals are ESI 2 vs. 23.7% of car arrivals — EMS pre-selects sicker patients.
9	triage_vital_rr	−0.10	Respiratory rate is a core early-instability marker, tracked in the expected direction.
10	triage_vital_hr	−0.10	Tachycardia/bradycardia signals compensatory shock; ties RR as the next-strongest vital.

Where clinical intuition and correlation disagree is the interesting part: triage_vital_sbp (systolic BP) barely correlates with ESI at all (r = 0.001) despite being a textbook shock marker — worth a specific look in Week 6, since it may mean hypotension is being compensated for elsewhere in the vitals, or that BP alone is a poor discriminator in this population.

Appendices

Evidence for every figure above is included alongside this memo:

1. **figs/01–06_*.png**
2. **notebooks**/Week5_Tutorial2_Data_Profiling.ipynb (profiling, missingness, outliers),
3. **notebooks**/Week5_Tutorial3_Exploratory_Visualisation.ipynb (the six-plot dashboard).

Glossary

A reference index for clinical, technical, and research terminology used throughout this proposal. Intended for any reviewer — clinical or non-clinical — who needs a quick definition without breaking flow.

Clinical & Vital Sign Terms

Term	Meaning
ED	Emergency Department
ESI	Emergency Severity Index — the 5-level triage scale used at Mercer General (1 = immediate, 5 = non-urgent)
SBP	Systolic Blood Pressure — the peak pressure during a heartbeat (top number in a BP reading)
DBP	Diastolic Blood Pressure — the pressure between heartbeats (bottom number in a BP reading)
MAP	Mean Arterial Pressure — average pressure driving blood to organs across a cardiac cycle; calculated as (SBP + 2×DBP) / 3
GCS	Glasgow Coma Scale — score from 3–15, measuring level of consciousness
RR	Respiratory Rate — breaths per minute

FiO2	Fraction of Inspired Oxygen — the percentage of oxygen a patient is receiving (room air = 21%)
SpO2	Oxygen Saturation — percentage of haemoglobin carrying oxygen, measured via pulse oximeter
SBAR	Situation, Background, Assessment, Recommendation — a structured format for clinical handover communication

Healthcare Systems & Infrastructure Terms

Term	**Meaning**
EHR	Electronic Health Record — digital patient record system; Mercer currently has none for triage, hence the Phase 1 proposal
LMIC	Low- and Middle-Income Country — a World Bank income classification used in global health literature
SIDS	Small Island Developing State — a UN classification for small island nations facing shared infrastructure constraints
Boarding time	The gap between a disposition decision (e.g. "admit") and the patient physically leaving the ED, usually due to no ward bed being available

AI / Machine Learning & Risk Terms

Term	Meaning
AI	Artificial Intelligence
ML	Machine Learning
AUROC / AUC	Area Under the Receiver Operating Characteristic curve — a model performance metric from 0.5 to 1.0; higher is better
XGBoost	Extreme Gradient Boosting — a machine learning algorithm using an ensemble of decision trees, common for structured/tabular data
Rule-based classifier	A decision-making system using explicit if/then logic rather than a trained statistical model
PPV	Positive Predictive Value — the proportion of positive alerts that turn out to be true positives. Low PPV drives alert fatigue.
Automation bias	The tendency for a human user to over-trust or defer to an automated system's output, even when contradicted by their own judgment
Distribution shift	When a model's performance degrades because the data it encounters in deployment differs statistically from the data it was trained on
Human-in-the-loop	A system design principle requiring a human to review, confirm, or be able to override an AI-generated output before it takes effect

Research & Documentation Terms

Term	Meaning
DOI	Digital Object Identifier — a permanent link format for academic papers
APA	American Psychological Association — the citation style used throughout this proposal (7th edition)
Scoping review	A literature review mapping the breadth of existing research, without necessarily assessing study quality in depth
Systematic review	A more rigorous literature review following a defined search and inclusion protocol